Semantic Consistency for Assuring Reliability of Large Language Models

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Abstract

Large Language Models (LLMs) exhibit remarkable fluency and competence across various natural language tasks. However, recent research has highlighted their sensitivity to variations in input prompts. To deploy LLMs in a safe and reliable manner, it is crucial for their outputs to be consistent when prompted with expressions that carry the same meaning or intent. While some existing work has explored how state-of-the-art LLMs address this issue, their evaluations have been confined to assessing lexical equality of single- or multi-word answers, overlooking the consistency of generative text sequences. For a more comprehensive understanding of the consistency of LLMs in open-ended text generation scenarios, we introduce a general measure of semantic consistency, and formulate multiple versions of this metric to evaluate the performance of various LLMs. Our proposal demonstrates significantly higher consistency and stronger correlation with human evaluations of output consistency than traditional metrics based on lexical consistency. Finally, we propose a novel prompting strategy, called Ask-to-Choose (A2C), to enhance semantic consistency. When evaluated for closed-book question answering based on answer variations from the TruthfulQA benchmark, A2C increases accuracy metrics for pretrained and finetuned LLMs by up to 47%, and semantic consistency metrics for instruction-tuned models by up to 7-fold.

Introduction

In recent times, the adoption of large pretrained language models (LLMs) in next-generation automated workflows for natural language-based tasks has been on the rise. However, this increased usage has also brought concerns about the safety and reliability of these models into the limelight (Weidinger et al. 2022; Gupta et al. 2023). In the context of Natural Language Generation (NLG), it is essential to have a dependable LLM that produces semantically equivalent outputs when given semantically equivalent prompts. This property is known as *consistency*. Consistency is critical to guarantee the safety of LLMs, since it increases the certainty that an LLM can produce similar enough outputs when fed with semantically similar inputs. However, the extent to which current NLG approaches exhibit consistency and the methods to measure this consistency remains insufficient. To address this gap, we propose a novel framework for evaluating and ensuring the consistency of NLG. Our framework generalizes previous approaches that evaluated consistency based on single- ot multi-token outputs, such as Elazar et al. (2021), to encompass entire generated text sequences.

Our measure of consistency for NLG surpasses simple lexical measures proposed in prior studies, and captures genuine variations in prompts that convey the same semantic meaning but differ in their lexical representation. We empirically assess the effectiveness of this metric in evaluating consistency, considering various measures of semantic equivalence, LLM architectures, as well as paraphrasing and answer generation/decoding techniques.

We consider a closed book question-answering scenario where we want to ensure that the answers generated from an LLM in response to paraphrased versions of a question are semantically similar. To this end, we utilize in-context learning strategies—i.e. eliciting answers from a LLM using few-shot examples in a prompt template—in a number of ways. Advanced prompting strategies, such as Chain-of-Thought reasoning (Wei et al. 2023), are widely known to extract improved performance from LLMs, as well as help reduce harmful bias (Guo, Yang, and Abbasi 2022) and improve factuality (Si et al. 2023). Our findings show that prompting strategies can also be useful to ensure consistency in realistic paraphrasing situations. We illustrate the specifics of our strategy in Fig. 1. To generate answers to paraphrased versions of a question, we start with the prompt template `paraphrasePrompt` that contains examples of different ways to paraphrase (such as using synonyms or changing syntax). We then iteratively feed the indicators of each of these methods, along with an input question into an auxiliary LLM (`AuxLLM`) to generate paraphrases of that question. These questions are fed into the main `LLM`, generating descriptive answers. Each descriptive answer, along with the original question, is added into a second prompt template (`answerPrompt`), and fed into `LLM` again to elicit short one or two word answers. The inclusion of few-shot examples of question-long answer-short answer combinations within the `answerPrompt` aid in this process. Once we have multiple short answers to the original question, we add each pair of answers, plus the original question, into `AuxLLM` again to obtain pairwise semantic equivalence of these two answers.

Our proposed semantic consistency metric takes as input all pairwise semantic equivalences, and calculates overall consistency between all variations of answers to the origi-

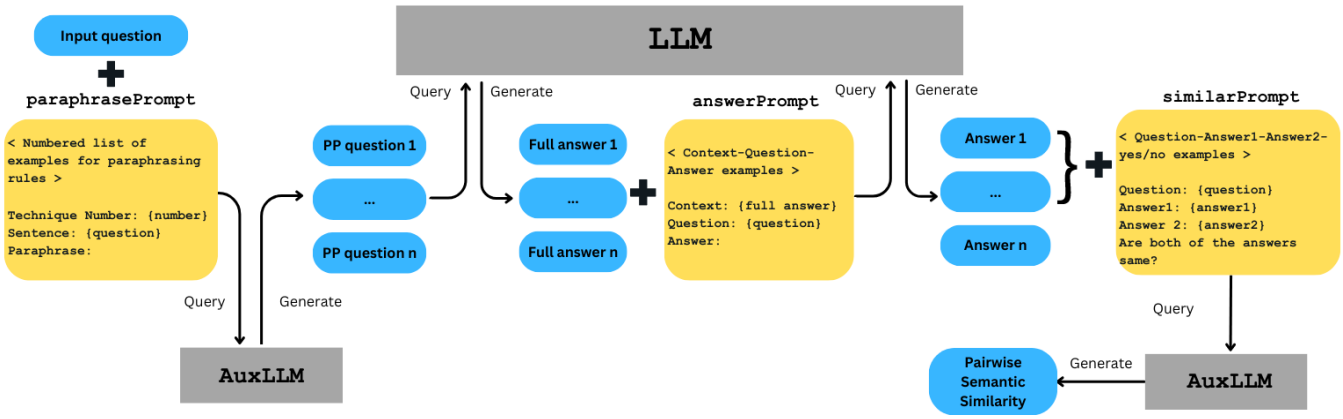

Figure 1: Illustration of the in-context learning pipeline for paraphrase generation and semantic similarity scoring.

nal question. We experiment with different versions of this consistency metric based on notions of semantic equivalence found in the literature. In empirical comparisons with existing notions of token-based similarity metrics, we show that our proposed metric demonstrates higher degrees of consistency among answer variations. Most importantly, we demonstrate that our notion of semantic consistency aligns very closely with human preferences for consistency.

Our main findings are as follows.

- Larger LLMs tend to exhibit higher levels of consistency.
- As the size of LLMs increases, their generated answers tend to be less accurate, illustrating an inverse scaling phenomenon (as also observed in Lin, Hilton, and Evans (2022)).
- Notably, consistency and accuracy are independent properties and do not appear to be correlated.
- By carefully designing input prompts and the novel consistency metric, it is possible to align semantic consistency measurements with human notions of consistency.
- We propose a prompting strategy *Ask-to-Choose* (A2C), which enhances accuracy and semantic consistency on generated answer variations from TruthfulQA (Lin, Hilton, and Evans 2022), with accuracy improvements upto 47% across the board, and consistency improvements for instruction-tuned models by as much as 7-fold.

Related Work

The concept of consistency measurement was first introduced in the LAMA probe to understand LLMs as knowledge bases (Petroni et al. 2019). Building on this idea, Elazar et al. (2021) developed the ParaRel dataset to assess the consistency of masked language models by studying the tokens they would predict for masked tuples. Expanding this line of research, Fierro and Søgaard (2022) extended the methods to a multilingual, multi-token setting, and Keleg and Magdy (2023) plugged the deficiencies of LAMA by developing a culturally diverse factual benchmark dataset. Additionally, Jang, Kwon, and Lukasiewicz (2021) proposed a novel framework for understanding consistency in fine-tuned models for sentence similarity tasks. Zhou et al. (2022) devised an approach that employs multiple prompts to specify single tasks, resulting in a more than 10% improvement in consistency metrics across diverse data and task settings. Finally, Newman, Choubey, and Rajani (2022) and Tam et al. (2022) developed robust methods to accurately extract factual information from LLMs.

Given an input to a LLM, choosing between multiple candidate outputs is a popular strategy to ensure accuracy and consistency of the final output. Among others, the Chain-of-Thoughts approach (Wei et al. 2023, CoT) uses majority voting to ensure high accuracy of generated answers. Kassner et al. (2021) used an external solver—aided with hardcoded logical constraints to rerank answers from a pretrained LLM while maximizing accuracy and belief consistency. Mitchell et al. (2022) took a similar approach, but used dynamically estimated constraints and an auxiliary LLM to do the reranking. Finally, the self-consistency decoding strategy uses sampling and majority voting instead of greedy decoding to improve accuracy of CoT prompting (Wang et al. 2022; Aggarwal et al. 2023). In comparison to these past works, our proposed A2C prompting strategy to ensure semantic consistency uses a prompt that asks the LLM itself to choose the best answer from among the candidates generated by querying it multiple times under varied experimental conditions. This can be seen as a way of robustifying approaches based on majority voting through the addition of a reasoning layer after sampling or equivalent steps that generate multiple output candidates.

We conclude by mentioning our previous work that motivates this paper. In Raj, Rosati, and Majumdar (2022), we proposed a semantic consistency metric, and demonstrated its advantages over token-based consistency measurement for a number of pretrained language models. In this paper, we significantly generalize the measurement framework beyond pairwise similarity, use in-context learning for paraphrase and output generation, and perform empirical evaluation on LLMs that are finetuned on instructions or with Reinforcement Learning from Human Feedback (RLHF). Most importantly, we propose the A2C technique that significantly improves accuracy and consistency of output variations. Note that while Raj, Rosati, and Majumdar (2022)

used three pretrained LLMs followed by a two-step filtering process to generate paraphrased questions, the obtained paraphrases were not diverse enough. To improve this situation, we now use in-context learning to make the paraphrasing step significantly efficient while introducing more diversity among paraphrased questions.

Methods

Consistency in LLMs refers to the expectation that similar prompts (such as paraphrases of each other) should generate similar outputs. This notion was first formalized by Elazar et al. (2021). Consider n *semantically similar* prompts $X = \{x_1, \ldots, x_n\}$ that generate outputs from an output space $\mathcal{Y}$: $Y = \{y_1, \ldots, y_n\}, y_i \in \mathcal{Y}$ when passed through a LLM. Given this, Elazar et al. (2021) defined consistency as

$$\text{Cons}_{lex}(Y) = \frac{1}{n(n-1)} \sum_{i,j=1, i \neq j}^{n} \mathbb{I}(y_i = y_j). \quad (1)$$

The above definition, as well as follow-up studies leveraging it (Fierro and Søgaard 2022; Jiang et al. 2020; Zhou et al. 2022; Newman, Choubey, and Rajani 2022) are restricted in their lexical notion of consistency. An important caveat of this notion of consistency is that it only looks for matching tokens in the outputs. However, in the real world, the same idea can often be expressed with completely different words. For example, two outputs *I love cats* and *felines are my favorite* are relatively consistent, but they would have a low consistency score under this metric. Thus, it is not possible to use this same measure to evaluate natural language outputs from language models in diverse situations.

Semantic Consistency

We propose a general measure of *semantic consistency* that substantially broadens the definition of Eq. (1) by substituting the lexical equality and simple average in Eq. (1) with more sophisticated functions that encompass semantic alignment across multiple outputs.

$$\text{Cons}_{sem}(Y) = \frac{1}{n} \sum_{i=1}^{n} S(y_i, Y_{-i}). \quad (2)$$

where $S(\cdot, \cdot)$ measures the extent of semantic similarity between each generated output y_i with the rest of the generated outputs, denoted by the set $Y_{-i} = \{1, \ldots, y_{i-1}, y_{i+1}, \ldots, y_n\}$. Note that the consistency measure of Elazar et al. (2021) is a special case of Eq. (2), taking $S(y_i, Y_{-i}) = (n-1)^{-1} \sum_{j \neq i} \mathbb{I}(y_j = y_i)$.

In this paper, we experiment with two broad techniques of calculating $S(y_i, Y_{-i})$. The first method is based on pairwise similarity. We simply replace the token equality indicator in Eq. (1) with a semantic equivalence function.

$$S(y_i, Y_{-i}) = \frac{1}{n-1} \sum_{j=1, i \neq j}^{n} s(y_i, y_j). \quad (3)$$

We can easily recover the original metric of Elazar et al. (2021) by setting the token equality indicator as $s(\cdot, \cdot)$, i.e. $s(y_i, y_j) \equiv \mathbb{I}(y_i = y_j)$. In our experiments, as the function $s(\cdot, \cdot)$ we use existing measures of semantic similarity, such as entailment and contradiction (Wang et al. 2018).

Our second instantiation of the semantic consistency metric in Eq. (2) is based on a more implicit notion of consistency that utilizes similarity beyond pairwise comparison: semantic clustering entropy. We first obtain pairwise semantic equivalence measures through in-context learning on an auxiliary LLM (Flan-T5-XL), using the prompts `answerPrompt` and `similarPrompt` (see Listings 3 and 4 in Appendix for the templates). We then use these pairwise scores to cluster the n outputs into $k \leq n$ semantically similar clusters, say $\mathcal{C} = \{C_i, \ldots, C_k\}$. Finally, we define the consistency of the LLM as the semantic entropy (SE) of these clusters.

$$SE = -\sum_{i=1}^{k} \left(\frac{|C_k|}{n} \right) \log_2 \left(\frac{|C_i|}{n} \right). \quad (4)$$

Eq. (4) adapts the information-theoretic definition of entropy over the discrete probability distribution where an element from $\mathcal{Y}$ is assigned to cluster C_k with probability $|C_k|/n$. Rearranging the terms in Eq. (4) with the definition in Eq. (2) in mind, we have as semantic equivalence function the following:

$$S(y_i, Y_{-i}) = -\sum_{k=1}^{K} (y_i \in C_k) \left[|C_k| \log_2 \left(\frac{|C_i|}{n} \right) \right].$$

Kuhn, Gal, and Farquhar (2023) utilized semantic clustering to determine the uncertainty in generations from a LLM, using a entailment-contradiction model as $s(\cdot, \cdot)$ to calculate the similarity score in both directions (i.e. $s(a, b) and s(b, a)$). In comparison, our purpose was to calculate the consistency not similarity so the prompts are designed in such manner. Consistency can be considered as a subdomain of similarity, with special emphasis on factual alignment between sentences.

Sources of Variation

To implement the semantic consistency measures outlined above, we first generate multiple paraphrases of a given prompt. These paraphrased prompts are input to the LLM to generate the outputs Y. We consider two different strategies for this process:

In-context Paraphrasing and Nucleus Sampling Realistic paraphrases are not completely random perturbations of the original phrase. Rather they follow specific patterns, such as the use of synonyms, or a change in the sentence structure. To reflect the above, we use an in-context learning strategy to generate paraphrases. We simply prompt an auxiliary LLM (text-davinci-003 (Brown et al. 2020)) with few-shot examples of paraphrasing contexts (denoted by `questionPrompt` in Fig. 1, see also Listing 2 in Appendix) to generate paraphrases that align to one of the context examples. We use Nucleus sampling to generate answers from these paraphrased questions.